THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of force is: 1

(a) newton **(b)** kilogram **(c)** joule **(d)** pascal

2. Hitting a moving ball with a bat changes its: 1

(a) mass **(b)** colour
(c) direction of motion **(d)** temperature

3. The force that opposes the motion between two surfaces is: 1

(a) buoyancy **(b)** gravity **(c)** friction **(d)** magnetism

4. Which is a non-contact force? 1

(a) push by hand **(b)** gravitational force
(c) friction **(d)** muscular force

5. Charges that do not move by themselves are called: 1

(a) current **(b)** static charges **(c)** fuses **(d)** magnetic poles

6. The weight of an object is the: 1

(a) amount of matter in it **(b)** force with which Earth pulls it
(c) density of it **(d)** volume of it

7. A body of mass 1 kg has weight about ___ on Earth. 1

(a) 1 N **(b)** 1.6 N **(c)** 10 N **(d)** 100 N

8. The range of a spring balance is the: 1

(a) length of the spring **(b)** minimum weight it can measure
(c) maximum weight it can measure **(d)** size of its hook

9. An object floats when gravitational force is: 1

(a) less than zero **(b)** equal to buoyant force
(c) greater than buoyant force **(d)** zero

10. An empty bottle pushed into water bounces back because of: 1

(a) static charge **(b)** upthrust **(c)** friction **(d)** magnetism

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define friction. State its direction. 2

12. Why do objects stop sooner on a rough road than on a smooth road? 2

13. State two properties of magnetic force. 2

14. What happens when two objects with like charges are brought near? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Compare magnetic force, electrostatic force and gravity in a table. 3

16. Describe the vertical motion of a ball thrown upwards. 3

17. Mass of an astronaut is 60 kg. Find weight on Earth and on the Moon (1.6 N/kg, Earth 10 N/kg). 3

OR

A body weighs 50 N on Earth. Find its mass. (1 kg = 10 N) 3

18. A spring balance has range 0–10 N with 1 N divided into 10 divisions. Find the least count. A load stretches it to 3 divisions after 4 N. What is the reading? 3

OR

Why do we say ‘weight of the wheat bag is 10 kg’ is not scientifically correct? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain how a spring balance works and how to read its least count, with an example. 5

OR

Explain floating and sinking in terms of forces. State Archimedes’ principle and give two applications. 5

20. Describe the activity of rubbing a balloon and its effects, and explain with electrostatic force. 5

OR

Explain why friction is both useful and harmful. Give four examples and two ways to reduce it. 5

END OF QUESTION PAPER — Total: 40 Marks

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